Modelling Incentives for Email Blocking Strategies

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Abstract

We consider the current arrangements whereby many users send their email via ISP "smarthosts" and model this system as an economic network. Some of these users may be insecurely configured so that their machines are hijacked for the sending of spam. This may in turn lead to some ISPs reducing spam loads by blocking all email from particular smarthosts. We derive formulae that express when blocking is an appropriate action and show that the number of legitimate users at the ISP must be taken into account as well as the number of spam sources. We relate this formal model to real world behaviour and attitudes and generate some strategies that maximise utility for different sizes of ISP. We present some quantitative email data from a large ISP that indicates the likely range in which numerical solutions will lie. We also demonstrate that some of the present approaches to email blocking are irrational and counterproductive.

1 Introduction

Many ways of filtering out unwanted bulk email (spam) have been proposed [9]. Content-based methods are often effective, but computationally expensive when done at the Internet Service Provider (ISP) [1]. A cheap and reasonably effective method of filtering is to simply refuse email from particular sources on the basis that they send nothing but spam. Many people delegate the blocking of spam to their ISP, sometimes without actually knowing that they are doing so [3]. Some ISPs compile their own blacklists to suit their own purposes, but many delegate their decisions still further by adopting the recommendations of public

blacklists, such as those from Spamhaus [11]. There are now well over 600 "blacklists"[8] that can be consulted when making blocking decisions, though the reasons given for inclusion on these lists varies widely. In late 2004 it was alleged that Verizon Inc had blacklisted all mail servers outside the USA, an action that has led to court action [6]. It is the use of blacklisting as a spam filtering (blocking) tactic and, in particular, the criteria being applied by ISPs that is the subject of this paper.

We argue that each ISP should add hosts to a blacklist only after having given due consideration to the incentives for doing so. We build a very simple model which allows us to show how this should be done, and present some evidence (based on interviews with Spamhaus and with Demon Internet, a large ISP in the UK) that something resembling this model takes place in the real world. Our model highlights some particular features of the email blocking problem, which, although rather obvious have not previously been clearly identified. We believe that our model can assist ISPs in *rationally* deciding whether to block a particular host or not.

2 The Model

Following the approach presented in [5], we model the email network as a graph with a set of nodes N and a set of edges $E \subseteq N \times N$. Nodes in this graph represent hosts that send out email – these could be ISP smarthosts (used by ISP customers as a relay for their outgoing email) or corporate email servers (that send their email directly, bypassing their ISP's systems).

Each node sends email on behalf of a collection of users, who are vulnerable to attacks from viruses, worms, trojan horses, etc. and hence may be unwittingly sending out spam. Of course there will also be some nodes that are owned by spammers and hence their sending of spam is entirely intentional – but only one third of all spam is now sent directly from spammers, and this proportion is rapidly decreasing [2].

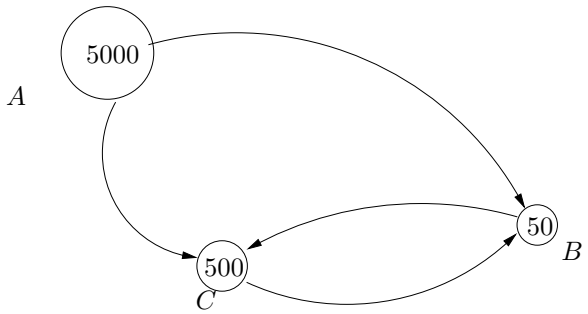


Figure 1: An Email Network

An example of an email network is shown in Figure 1. In the diagram the arrows express the flows of emails which are permitted within the network.

Hence in this example, the large ISP A is not happy to receive email from either B or C, whereas they are both prepared to swap email with each other and accept it from A.

The utility of each of the hosts of the network is a function of its connectivity (who email can be sent to and who it can be received from) and of the amount of spam being received from remote sites. Clearly any particular user is happiest if she is able to send email to and receive email from as many people as possible, while receiving the least possible amount of spam.

We use C_A to denote the set of users (clients) belonging to node A and a function $V_A : C_A \to \mathbb{N}$ to represent the cost of receiving the spam sent by A's users (due to intentional sending, or because some of them have been compromised by the various attacks) to one random email address. Essentially, V_A measures how vulnerable the users C_A are. We can then express the utility of a node as follows:

$$U(A) = \left(\sum_{B \text{ s.t. } (A,B) \in E} U(C_B) \right) - \left(|C_A| \times \sum_{B \text{ s.t. } (B,A) \in E} V_B(C_B) \right)$$

where the first, positive, term involves $U(C_B)$, the utility of being able to send email to the clients of node B and the second, negative, term involves $V_B(C_B)$, the vulnerability of the clients C_B (the likelihood they will send spam) which we then multiply up by all the clients C_A of A that they might send it to.

However, it is not only the vulnerability of clients which might cause A to block the flow of email from B. If the incentives of node A (the ISP, the smarthost, the corporation running the email server) are different to those of $a \in C_A$, i.e. the users of the smarthost), then blocking could take place contrary to the wishes of the users. One such example is corporations blocking email flow to and from free email services such as Hotmail or Gmail due to "too much personal use of company resources". The authors recently came across one case of this, and believe it will become commonplace.

One might expect to see a further positive term in the formula for $U(A)$ to express the utility of being able to receive email from remote sites, but we do not include it. The reason is that there is significant anecdotal evidence suggesting that whether users are able to receive email or not is irrelevant to the overall utility function. Users rarely learn about the email they do not receive, and even if they do, are not competent enough to assign the blame to their ISP blocking it rather than the sender having made an error.

Of course, a great deal of email will not actually travel to remote ISPs but will be delivered locally. ISPs tend to serve small geographical areas and this, along with their market positioning, will mean that they have similar types of clientele who are likely to exchange email amongst themselves. Also, the very large global entities, such as AOL or MSN, will send a lot of email internally because of their sheer size. We can therefore expect the self-sending terms of the utility function:

$$U(A)_{self} = U(C_A) - |C_A| \times V_A(C_A)$$

to be especially large. These terms may be even larger because each ISP will be in an excellent position to act against their own users sending spam to each other and hence the negative term will tend to zero. So we can expect the largest term in the utility function to be:

$$U(A)_{self} = U(C_A)$$

One might reasonably argue that summing utility values is merely a lower bound to the total utility and that one might expect to see a power function (following "Metcalf's Law", this would be squaring[1]). We will revisit this later in the paper.

Of course, each of the nodes has complete control over whether it will receive email from other each of the nodes, i.e. over the edges coming from it. Hence, if nodes are behaving rationally they will accept email from all nodes where this increases overall utility, viz: node A should have an edge to node B when $U(C_B) - |C_A| \times V_B(C_B) \geq 0$. Quite clearly, where the "vulnerability" is complete (viz: the node is owned by spammers and sends nothing but spam) then it is obvious that the utility is negative and blocking is the correct strategy.

3 Implications of the Blocking Model

The model shows one way to rationally decide whether to block email from a particular node or not. This has several implications, which we relate to situations which have arisen in practice.

First of all, ISPs are adversely affected by their customers' security vulnerabilities. Some ISPs already recognise this and take a proactive approach, scanning their customers' machines for problems and notifying them that patches need to be applied. This is expensive, but it reduces the possibility of customers' machines being compromised (represented in our model by V_A), spam being seen to come from the ISP's smarthost, and the smarthost being blocked.

Clearly, the model also implies that "secure" customers are more valuable than insecure ones. For instance, customers running operating systems and email clients which are not vulnerable to standard attacks are much less likely to be compromised and thereby cause spam to be sent through the ISP smarthost. In practice it is the presence of insecure or actively spamming customers that it usually acted upon by other ISPs. According to reports [7], a high percentage of traffic coming from one large UK ISP constitutes "advanced fee fraud" scams, which causes all of its email to be blocked by many other ISPs. Another case was that of a large US ISP which never terminated accounts used by spammers. Hence, many of its smarthosts became blocked and customers were unable to send email (interestingly, the company policy only changed when the corporate email server was blocked, thus preventing executives from sending email!).

Spam is sent in two distinct ways, scattershotted to a selection of valid email addresses, or as an indiscriminate "dictionary attack" that attempts to

[1] "the power of the network, how much it can do, is the square of the number of connected machines", Robert Metcalf, 1973

find deliverable addresses by random guessing. In the latter case, the size of receiving ISP is irrelevant, but in the former case, the larger the ISP, the more likely it is to receive spam and the more likely it is that there will be users who notice. This means that there will be a tendency for larger ISPs to be far more aware of spam sources and hence they will tend to block more. This seems to be supported by evidence in practice (Demon seems to be regularly blocked by one careless large US ISP [4]). In particular, since the self-sending term is so large, almost all the other terms will be "lost in the noise" and it may be that many users are entirely, or almost entirely, unaware of extensive blocking. Conversely, a small ISP will see little benefit from blocking and its utility function will be dominated by the ability to send to others.

We have argued that utility is unaffected by whether or not users receive email, this could mean "tit for tat" blocking is a useful strategy. If A has decided to block all email arriving from B then B might consider blocking incoming email from A in retaliation on the basis that the users C_B will not notice. However, whether A takes any notice will depend on the size of the two terms in the model, which will significantly affected by the relative sizes of the two entities. If C_A is large compared with C_B then A may just not care that they have been blocked, however if B is the larger entity then it may be that "tit for tat" will be effective in getting the block removed.

Demon felt that their reputation was of considerable importance. viz: that other ISPs would not block them immediately but would give them a chance to fix things because they had managed such fixes in the past. This corresponds to the other ISPs being unable to measure $V_B(C_B)$ but relying on estimates of its likely size. An obvious corollary of this is that it is important for an ISP to be publicising its effectiveness at dealing promptly with spam coming from its systems. This makes it less likely to be blocked because of the over-optimistic view that others will then take of the vulnerability term.

Demon also told us that email blocking was often long term. Some sites would be quick to block and would take a significant time to unblock even when the original problem had been fixed long before. This is irrational because it uses an overly pessimistic view of the vulnerability term, but our simple model does not include any costs for making decisions about blocking and implementing them; and here we are probably seeing a lack of resources for making changes.

4 Experimental Results

We were able to examine four weeks data from the Demon smarthost. We excluded *nnn* customers who, during this period, were detected to be infected with viruses or worms, or those who were insecure and were being exploited to send spam. The data covers 24,123,456 emails which were addressed to 47,123,456 distinct destinations. We found, etc [firm results are still in progress . . . but we expect to say something more about Metcalf's views on connectivity values]

5 Conclusion

In this paper we have outlined an initial approach to understanding the economics of using email blocking to combat spam. We pursued two directions – designing a simple model for this problem and trying to find some evidence supporting it by conducting interviews at a UK ISP and Spamhaus.

From this we gained an insight into this very practical problem, and this has enabled us to make interesting observations of what is happening now and how things might change in the future.

Our access to some quantitative data means that we have been able to make some suggestions as to the shape of the utility function and hence we were able to provide further illumination.

We are not yet able to make clear recommendations as to whether it is rational for ISP A to block ISP B's smarthost. However, by reporting some of the most interesting observations arising from our work so far we hope to stimulate further research in this area.

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